

Using Localized Facial Regions for Robust rPPG Measurement in Obstructed Visibility Environments

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Abstract

In immersive virtual reality (VR) environments, real-time non-contact heart rate (HR) monitoring can enhance user experience by supporting adaptive system responses and enabling more natural human-computer interaction. While remote photoplethysmography (rPPG) has emerged as a promising technique for unobtrusive physiological measurement, conventional facial regions of interest (ROIs) such as the forehead and cheeks are often compromised by motion artifacts, skin tone variations, makeup, and occlusion from head-mounted displays (HMDs). This study explores the potential of the ear region as a reliable alternative ROI for rPPG-based HR estimation. This study systematically evaluated ear-based rPPG performance across seven experimental conditions, including variations in lighting, facial makeup, and task dynamics, using a controlled VR setup with synchronized ground-truth PPG recordings. The findings indicate that the ear region consistently exhibited superior performance in comparison to the cheek region, as evidenced by reduced mean absolute error (MAE; e.g., 1.23 bpm vs. 3.11 bpm) and enhanced Pearson correlation (e.g., $r = 0.81$ vs. $r = 0.50$ under makeup conditions), culminating in an aggregate pooled correlation of $r = 0.97$ under baseline conditions, with respect to ground-truth HR derived from synchronized PPG recordings. These findings demonstrate the practical viability of integrating ear-based rPPG into next-generation VR systems for continuous, unobtrusive physiological monitoring.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI); Ubiquitous and mobile computing.

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Keywords

remote Photoplethysmography (rPPG), virtual reality, heart rate monitoring

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1 Introduction

In immersive environments such as virtual reality (VR), real-time physiological monitoring is becoming increasingly essential for ensuring user safety, optimizing experiences, and enabling adaptive system responses [13, 14]. Conventional methods such as electrocardiography (ECG) [12] and contact-based photoplethysmography (PPG) [3] are widely employed; however, these methods necessitate direct skin contact, frequently resulting in discomfort, motion constraints, and practical limitations, particularly during extended use or in cases of sensitive skin [2]. Although certain areas of the VR headset, such as the forehead or straps, could support contact sensors, their integration often requires conductive materials and precise placement, potentially compromising comfort and design flexibility. To address these limitations, remote photoplethysmography (rPPG) has emerged as a promising non-contact alternative. By analyzing subtle skin color variations captured by standard video cameras, rPPG enables unobtrusive heart rate monitoring [9]. Moreover, as many VR headsets already include embedded RGB cameras, rPPG can be implemented with minimal hardware modification.

While rPPG has demonstrated considerable potential as a non-contact physiological monitoring technique, the majority of existing research has concentrated on facial regions, particularly the forehead and cheeks, as regions of interest (ROIs) [8]. However, these ROIs are highly susceptible to real-world factors such as facial expressions, skin tone variations [17], makeup [10], and occlusions from face masks or head-mounted displays (HMDs). These factors

introduce motion artifacts and optical inconsistencies that compromise signal quality. To advance practical rPPG implementation, it is crucial to identify and validate alternative ROIs with greater resilience under such challenging conditions.

In addressing the limitations of conventional facial ROIs, this study proposes the ear region as a promising alternative for rPPG signal extraction. In contrast to the regions of the face such as the cheeks or forehead, the ear exhibits minimal sensitivity to facial expressions and variations in skin tone, and it remains largely unobstructed even when users wear HMDs or face masks, which are common in VR environments. A systematic investigation was conducted to assess the feasibility and robustness of ear-based rPPG. This investigation involved an experimental comparison of ear-based rPPG with cheek-based measurements under immersive VR conditions. The performance is evaluated across a range of realistic scenarios, encompassing variations in lighting, makeup, and dynamic versus static movements. This study makes two key contributions: (1) it provides empirical evidence for the ear as a stable and reliable ROI for rPPG and (2) it offers a comprehensive performance evaluation across diverse real-world conditions, advancing the practical deployment of rPPG technologies.

2 Method

2.1 Participants

For the experiment, fifteen healthy adult volunteers (8 males, 7 females, mean age (SD) 23.19 (2.75) years) participated. Following the provision of informed consent, participants were fitted with a Meta Quest Pro HMD (Meta, Menlo Park, CA, USA), and wore a Polar Verity Sense (Polar, Kempele, Finland), a wrist-worn PPG sensor, on their dominant wrist. The study was approved by the Institutional Review Board (IRB) of Sookmyung Women's University (SMWU-2405-HR-019).

2.2 Materials

2.2.1 Dynamic VR Contents. To evaluate the reliability of HR measurements under dynamic conditions, the experiment incorporated interactive VR games from the Meta Store that required participants to actively use controllers and engage in whole-body and head movements. Two distinct one-minute scenarios were presented: a dark-toned haunted mansion exploration [7] and a bright-toned downhill snowboarding experience on a snowy mountain [5]. Participants stood upright and were permitted to move freely, simulating realistic, high-motion VR interactions.

2.2.2 Static VR Videos. To assess HR measurement performance under static conditions, the experiment employed 360-degree VR videos sourced from YouTube. Participants were seated and instructed to passively observe the virtual environments without controller interaction, though they were allowed to slowly rotate their heads. Two distinct one-minute scenarios were used: a dark-toned panoramic view of a starry night sky [15] and a bright-toned river rafting experience [18]. These sessions minimized body movement, allowing isolation of the effects of environmental brightness and visual stimuli on rPPG signal extraction.

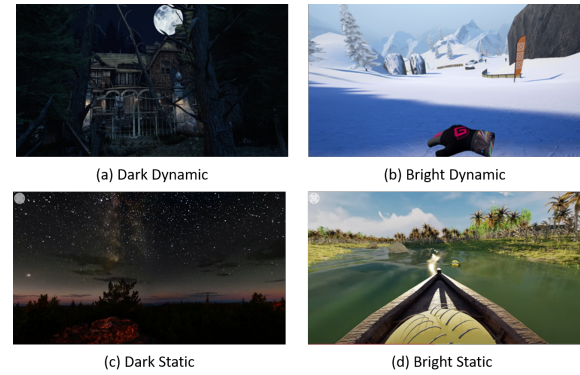


Figure 1: Still frame of VR contents and video clips used in the experiment for dark/bright and dynamic/static condition. (a) Dark dynamic condition; (b) Bright dynamic condition; (c) Dark static condition; (d) Bright static condition;

2.3 Hardware Setup and Recording Devices

The Meta Quest Pro HMD facilitated immersive VR experiences while naturally occluding conventional facial regions, such as the forehead and cheeks, thereby creating realistic conditions for evaluating alternative rPPG sites. Two Raspberry Pi Camera Module 3 Wide units (Raspberry Pi Ltd., Cambridge, UK) were mounted on either side of the HMD using clips and wire, with each unit targeting the cheek or ear region at 4608×2592 pixels and 30 frames per second (fps). Ground-truth HR was recorded via a Polar Verity Sense wrist-worn PPG sensor on the participant's dominant hand. The experimental environment was meticulously controlled to guarantee consistent camera alignment, stable lighting, and minimal external interference, thereby facilitating robust, synchronized acquisition of video and physiological signals across all conditions.

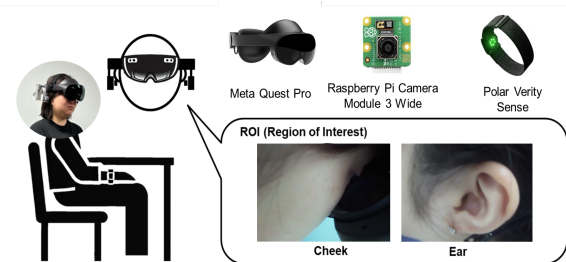


Figure 2: Experimental hardware configuration and recording setup. The figure illustrates the hardware components and setup used for simultaneous rPPG and ground-truth HR data acquisition.

During data acquisition, the Raspberry Pi on the HMD annotated each video frame with a normalized timestamp at one-second intervals via Wi-Fi, while the Polar device timestamped heart rate data transmitted over Bluetooth Low Energy (BLE). After the experiment, rPPG signals extracted from the video were temporally aligned with the Polar HR data using these timestamps, enabling accurate comparison and analysis.

2.4 Experimental Design

The experiment assessed the robustness of rPPG signals under seven conditions, varying in lighting, makeup application, and dynamic versus static VR interactions. Prior to the commencement of the primary tasks, the participants underwent a preparatory phase that entailed instructions, equipment configuration, and a familiarization session with the VR system. Each participant completed seven one-minute trials, separated by three-minute rest periods, for a total duration of approximately 40 minutes. The condition order was randomized to minimize order effects. Specifically, the baseline involved sitting quietly with eyes open; the room light condition involved turning off room lights; and the makeup condition involved applying foundation to the face. In dynamic tasks, participants engaged with VR games while standing; in static tasks, they remained seated and watched 360-degree videos. To investigate whether the color tone of the VR content (bright or dark) affected rPPG signal quality, both dynamic and static tasks incorporated VR scenarios designed to modulate the amount of light reflected onto the cheek through the HMD. Throughout the course of all trials, synchronized video recordings of the subjects' cheeks and ears were collected, as well as ground-truth HR data, which were then used for the purpose of performance comparison.

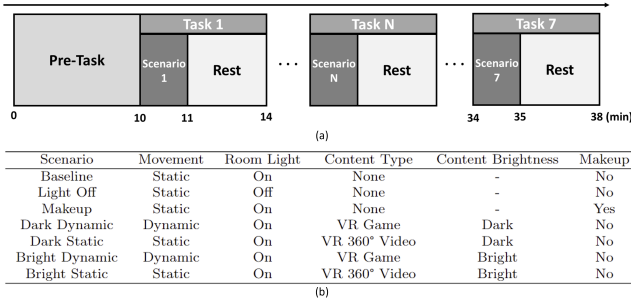


Figure 3: Experimental paradigm and task conditions. (a) Experimental paradigm; (b) Summary detailing the characteristics of each experimental condition;

2.5 Signal Recording and Processing

Ground-truth HR data were recorded using the Polar Verity Sense, a wrist-worn PPG sensor capable of providing continuous HR monitoring at a frequency of one measurement per second. Simultaneously, facial videos were captured using two side-mounted Raspberry Pi cameras, with one dedicated to recording the cheek region and the other dedicated to recording the ear region. All signal processing and analysis were implemented in Python (version 3.11.12) using OpenCV (version 4.11.0).

2.5.1 rPPG Processing. The rPPG extraction and pulse rate estimation procedures were based on the method described by De Haan et al. [6]. From the recorded video frames, RGB signals were extracted separately from the defined ROIs for the cheek and ear, and skin masks were applied to isolate skin pixels. A chrominance-based approach was employed to generate the pulse signals, computing $X = R - G$ and $Y = 0.5R + 0.5G - B$, followed by the standardized

combination $S = X - \alpha Y$, where α is defined as the ratio of the standard deviations of X and Y ($\alpha = \sigma_X / \sigma_Y$). The resulting signals from both regions were bandpass-filtered (0.67–4 Hz) to isolate the heart rate frequency band. For HR estimation, a sliding window of at least 3 seconds (90 frames) with 2-second overlap was used to apply a Fast Fourier Transform (FFT), identifying the dominant frequency peak (f_{peak}) within the 0.67–4 Hz range, which was converted to beats per minute (BPM) as $HR = f_{\text{peak}} \times 60$. This setup enabled the parallel extraction and comparison of HR estimates from both the cheek and ear regions against ground-truth measurements.

3 Experimental Results and Analysis

This section presents experimental results evaluating the accuracy and robustness of rPPG-based heart rate estimation from the cheek and ear regions under varying environmental and task conditions. The analyses compare estimated HR to ground-truth data across interaction types, lighting conditions, and facial makeup, highlighting the ear's viability as an alternative ROI in VR settings where conventional facial areas are often occluded.

Table 1 summarizes the subject-wise average performance metrics for rPPG-derived HR estimation compared to ground-truth PPG measurements across seven experimental conditions. The reported metrics include the root mean square error (RMSE), mean absolute error (MAE), and Pearson correlation coefficient (r) for both the cheek and ear regions. Overall, the ear region consistently outperformed the cheek region, demonstrating lower error rates and higher correlation values across most conditions. Notably, under challenging scenarios such as the makeup condition, the ear region maintained robust performance (MAE = 1.23 bpm, $r = 0.81$) compared to the cheek region, which exhibited substantial performance degradation (MAE = 3.11 bpm, $r = 0.50$). Similarly, under dark dynamic conditions, the ear region showed superior resilience, achieving correlation coefficients exceeding 0.8, while the cheek region remained below 0.7. These results highlight the ear region's potential as a robust alternative ROI for rPPG-based HR estimation in real-world VR applications where conventional facial regions may be occluded or affected by environmental factors.

Scenario	RMSE		MAE		Corr	
	Cheek	Ear	Cheek	Ear	Cheek	Ear
Baseline	1.48 (0.13)	1.44 (0.13)	1.20 (0.13)	1.16 (0.13)	0.85 (0.02)	0.86 (0.02)
Light Off	2.42 (0.22)	2.39 (0.23)	1.94 (0.23)	1.92 (0.23)	0.71 (0.05)	0.72 (0.05)
Makeup	3.89 (0.45)	1.53 (0.11)	3.11 (0.38)	1.23 (0.12)	0.50 (0.09)	0.81 (0.05)
Dark Dynamic	2.74 (0.23)	1.57 (0.14)	2.23 (0.23)	1.25 (0.12)	0.60 (0.08)	0.82 (0.04)
Dark Static	2.43 (0.23)	1.60 (0.16)	1.96 (0.22)	1.28 (0.11)	0.68 (0.05)	0.83 (0.03)
Bright Dynamic	2.53 (0.24)	2.19 (0.22)	2.04 (0.23)	1.75 (0.16)	0.66 (0.05)	0.75 (0.04)
Bright Static	2.01 (0.20)	1.53 (0.11)	1.61 (0.14)	1.22 (0.10)	0.73 (0.06)	0.82 (0.03)

Table 1: Subject-wise average performance metrics for rPPG-derived HR estimation compared with ground-truth PPG measurements. The best value for each metric is highlighted in bold. For RMSE and MAE, lower values indicate higher accuracy, while for Corr, higher values indicate stronger correlation. All best-performing values were obtained under the baseline scenario using the Ear region.

Figure 4 presents the scatter plot designed to verify that HR values computed using the ear-based rPPG method closely approximate the ground-truth PPG measurements, thereby demonstrating the method's reliability under baseline conditions. The figure

reflects the pooled-data correlation between rPPG-derived HR estimates from the ear region and ground-truth HR values across all participants and trials. The fitted regression line indicates a strong positive association, with a Pearson correlation coefficient of $r = 0.97$ and statistical significance of $p < 0.001$, capturing the overall relationship aggregated across the combined dataset.

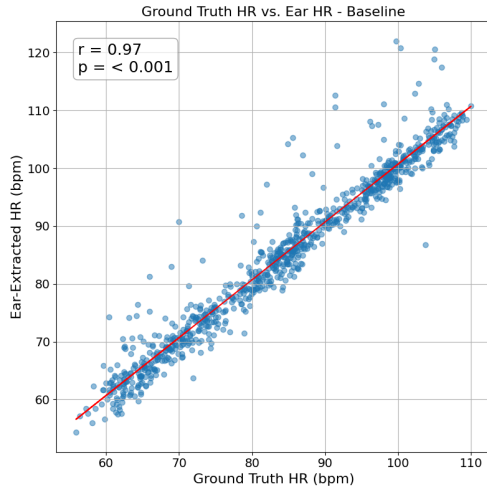


Figure 4: Scatter plot between pooled rPPG-derived HR and ground-truth HR across all participants under the baseline condition. The plot shows the correlation calculated across the combined dataset from all participants, with a Pearson correlation coefficient of $r = 0.97$ and statistical significance of $p < 0.001$.

To further examine the robustness of ear-based rPPG performance, we conducted condition-wise analyses comparing key experimental factors, including dynamic versus static interactions, bright versus dark VR contents, and the presence versus absence of facial makeup. Across these comparisons, the ear region consistently demonstrated superior performance relative to the cheek region. Specifically, under dynamic conditions, where participant movement and head motion introduced additional artifacts, the ear region maintained high correlation values (e.g., $r > 0.8$) and low error metrics, whereas the cheek region exhibited notable performance degradation. Similarly, under dark-content conditions, which reduced facial illumination and challenged traditional rPPG extraction, the ear-based measurements remained robust. Notably, the most pronounced performance disparity was observed under the makeup condition, where cheek-derived signals suffered substantial attenuation and increased noise, while ear-derived signals remained largely unaffected. These findings highlight the ear region’s resilience across diverse real-world conditions, underscoring its potential as a reliable alternative ROI for non-contact HR monitoring in immersive VR environments.

The experimental results demonstrate the robustness of ear-based rPPG for heart rate estimation across various challenging conditions, including motion, low illumination, and facial makeup. Compared to the cheek, the ear consistently yielded higher accuracy and lower error, particularly under dynamic and dark-content

scenarios. The most notable performance gap was observed in the makeup condition, where cheek signals degraded significantly while ear signals remained stable. These outcomes highlight the ear’s resilience to occlusion, motion artifacts, and cosmetic interference, supporting its suitability as a reliable ROI for non-contact HR monitoring in immersive VR environments.

4 Future Works and Conclusion

While the present study offers compelling evidence supporting the robustness and reliability of ear-based rPPG for heart rate (HR) estimation under diverse VR conditions, several limitations warrant acknowledgment. First, the participant sample was relatively small and homogeneous, potentially limiting the generalizability of the findings to broader, more diverse populations. Additionally, the experimental procedures were conducted within a controlled laboratory setting, which may not fully capture the complexities and environmental variability of real-world VR applications, such as fluctuating ambient lighting, diverse user behaviors, or heterogeneous hardware configurations. Future research should address these limitations by expanding the sample size to encompass a wider range of demographic and physiological characteristics and by validating the method in ecologically valid VR scenarios that reflect practical use cases. Moreover, the integration of advanced signal enhancement methodologies, including motion artifact reduction algorithms [1, 11] and adaptive filtering techniques [4, 16], holds substantial promise for further improving the resilience of ear-based rPPG, particularly under conditions of extreme motion or challenging illumination. Despite these limitations, the present findings highlight the significant potential of the ear region as an alternative ROI for non-contact physiological monitoring in immersive VR environments. The demonstrated reliability of ear-based rPPG across multiple challenging conditions positions it as a promising solution for continuous, unobtrusive health monitoring in next-generation VR systems, with broad applications in healthcare, training, gaming, and beyond.

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